

CASE

$X_1, X_2, \dots, X_n$ i.i.d.
 $E[X_i] = \mu$ $\text{Var}(X_i) = \sigma^2$


 θ
 μ

$\hat{\theta}$
 $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$
 SE: $\sigma/\sqrt{n}$

 $\hat{\theta}$ $\bar{x}$

se: standard error estimate

 $\frac{s}{\sqrt{n}}$

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

$$ci_{1-\alpha}(\mu) = \bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

confidence interval for μ
 with known variance σ^2

$$ci_{1-\alpha}(\mu) = \bar{x} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$$

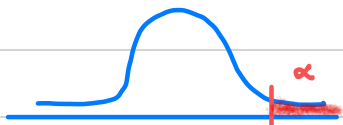
confidence interval for μ
 with unknown variance

Ⓟ `L1.mean(x, sigma, conf.level)`

PERVS XXX : PERVS 1

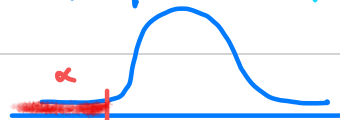
$$H_0: \mu \leq \mu_0 \text{ (or } \mu = \mu_0)$$

$$H_1: \mu > \mu_0 \text{ (or } \mu = \mu_1 > \mu_0)$$



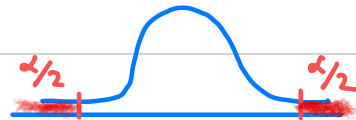
$$H_0: \mu \geq \mu_0 \text{ (or } \mu = \mu_0)$$

$$H_1: \mu < \mu_0 \text{ (or } \mu = \mu_1 < \mu_0)$$



$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0$$



KNOWN VARIANCE σ^2

$$\frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \stackrel{H_0}{\sim} N(0,1)$$

PIVOTAL QUANTITY

$$Z\text{-score} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \quad \text{REALISATION OF PIVOTAL QUANTITY}$$

$$R_\alpha = \{Z\text{-score} > \boxed{z_\alpha}\}$$

$q_{\text{norm}}(1-\alpha) \leftarrow$

$$\text{p-value} = P(Z > Z\text{-score})$$

$\leftarrow 1 - \text{pnorm}(Z\text{-score})$

$$R_\alpha = \{Z\text{-score} < \boxed{-z_\alpha}\}$$

$q_{\text{norm}}(\alpha) \leftarrow$

$$\text{p-value} = P(Z < Z\text{-score})$$

$\leftarrow \text{pnorm}(Z\text{-score})$

$$R_\alpha = \{Z\text{-score} < \boxed{-z_{\alpha/2}} \text{ or } Z\text{-score} > \boxed{z_{\alpha/2}}\}$$

$q_{\text{norm}}(1-\alpha/2) \leftarrow$

$$\text{p-value} = 2 \cdot P(Z > |Z\text{-score}|)$$

$\leftarrow [1 - \text{pnorm}(|Z\text{-score}|)] \cdot 2$

UNKNOWN VARIANCES

$$\frac{\bar{X} - \mu_0}{s / \sqrt{n}} \stackrel{H_0}{\sim} t_{n-1}$$

PIVOTAL QUANTITY

$$t\text{-score} = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} \quad \text{REALISATION OF PIVOTAL QUANTITY}$$

$$R_\alpha = \{t\text{-score} > \boxed{t_{n-1, \alpha}}\}$$

$q_{t(1-\alpha, n-1)} \leftarrow$

$$\text{p-value} = P(t_{n-1} > t\text{-score})$$

$\leftarrow 1 - \text{pt}(t\text{-score}, n-1)$

$$R_\alpha = \{t\text{-score} < \boxed{-t_{n-1, \alpha}}\}$$

$q_{t(\alpha, n-1)} \leftarrow$

$$\text{p-value} = P(t_{n-1} < t\text{-score})$$

$\leftarrow \text{pt}(t\text{-score}, n-1)$

$$R_\alpha = \{t\text{-score} < \boxed{-t_{n-1, \frac{\alpha}{2}}} \text{ or } t\text{-score} > \boxed{t_{n-1, \frac{\alpha}{2}}}\}$$

$q_{t(1-\frac{\alpha}{2}, n-1)} \leftarrow$

$$\text{p-value} = 2 \cdot P(t_{n-1} > |t\text{-score}|)$$

$\leftarrow [1 - \text{pt}(|t\text{-score}|, n-1)] \cdot 2$

CASE

 θ $\hat{\theta}$ $\hat{\theta}$

se: standard error estimate

 $X_1, \dots, X_n$ i.i.d.

2

 $X_i \sim \text{Bernoulli}(p)$ $E[X_i] = p, \text{Var}(X_i) = p(1-p)$ p

$$\hat{P} \sim N(p, \frac{p(1-p)}{n})$$

$$SE: \sqrt{\frac{p(1-p)}{n}}$$

 $\hat{p}$

$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \sim N(0,1)$$

 $\sim N(0,1)$

$$\rightarrow ci_{1-\alpha}(p) = \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

confidence interval for p
(variance always unknown)So that the ci has

$$ME \leq 0.025 \quad \text{width} \leq 0.05$$

⑤ $ci.prop(x, success, conf.level)$

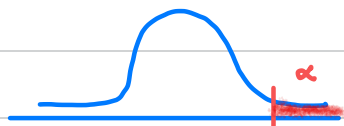
$$\frac{z_{\alpha/2}^2}{0.025^2} \hat{p}(1-\hat{p}) \leq \frac{z_{\alpha/2}^2}{0.025^2} 0.5^2 \leq n$$

at most 0.5^2

PERVS XXX : PERVS 2

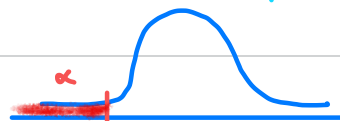
$$H_0: p \leq p_0 \text{ (or } p = p_0)$$

$$H_1: p > p_0 \text{ (or } p = p_1 > p_0)$$



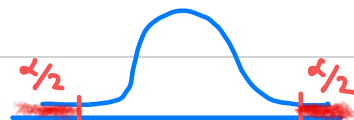
$$H_0: p \geq p_0 \text{ (or } p = p_0)$$

$$H_1: p < p_0 \text{ (or } p = p_1 < p_0)$$



$$H_0: p = p_0$$

$$H_1: p \neq p_0$$



$$\frac{\hat{P} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \stackrel{H_0}{\sim} N(0,1)$$

PIVOTAL
QUANTITY



$$Z\text{-score} = \frac{\hat{P} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

REALISATION OF
PIVOTAL QUANTITY

$$R_\alpha = \{Z\text{-score} > \boxed{Z_\alpha}\}$$

$q_{\text{norm}}(1-\alpha)$ ←

$$\text{p-value} = P(Z > Z\text{-score})$$

← $1 - p_{\text{norm}}(Z\text{-score})$

$$R_\alpha = \{Z\text{-score} < \boxed{-Z_\alpha}\}$$

$q_{\text{norm}}(\alpha)$ ←

$$\text{p-value} = P(Z < Z\text{-score})$$

← $p_{\text{norm}}(Z\text{-score})$

$$R_\alpha = \{Z\text{-score} < \boxed{-Z_{\alpha/2}} \text{ or } Z\text{-score} > \boxed{Z_{\alpha/2}}\}$$

$q_{\text{norm}}(1-\alpha/2)$ ←

$$\text{p-value} = 2 \cdot P(Z > |Z\text{-score}|)$$

← $[1 - p_{\text{norm}}(|Z\text{-score}|)] \cdot 2$

CASE

 θ $\hat{\theta}$ $\hat{\theta}$

se: standard error estimate

 $X_1, \dots, X_{n_x}$ i.i.d. $E[X_i] = \mu_x$ $\text{Var}(X_i) = \sigma_x^2$ $Y_1, \dots, Y_{n_y}$ i.i.d. $E[Y_i] = \mu_y$ $\text{Var}(Y_i) = \sigma_y^2$ x, y independent

$$s_{\text{pool}}^2 = \frac{(n_x - 1)s_x^2 + (n_y - 1)s_y^2}{n_x + n_y - 2}$$

 $\mu_x - \mu_y$
(μ_d)

$$\bar{X} - \bar{Y} \sim N(\underbrace{\mu_x - \mu_y}_{=: \mu_d}, \frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y})$$

$$SE: \sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}$$

 $\bar{x} - \bar{y}$ If $\sigma_x^2 \neq \sigma_y^2$:

$$\sqrt{\frac{s_x^2}{n_x} + \frac{s_y^2}{n_y}}$$

If $\sigma_x^2 = \sigma_y^2$:

$$\sqrt{\frac{s_{\text{pool}}^2}{n_x} + \frac{s_{\text{pool}}^2}{n_y}}$$

$$\frac{\bar{X} - \bar{Y} - (\mu_x - \mu_y)}{\sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}} \sim N(0,1)$$

 $\sim N(0,1)$

$$\rightarrow CI_{1-\alpha}(\mu_x - \mu_y) = \bar{x} - \bar{y} \pm Z_{\alpha/2} \sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}$$

confidence interval
for $\mu_x - \mu_y$,
 σ_x^2, σ_y^2 known

$$\frac{\bar{X} - \bar{Y} - (\mu_x - \mu_y)}{\sqrt{\frac{s_{\text{pool}}^2}{n_x} + \frac{s_{\text{pool}}^2}{n_y}}} \sim t_{n_x + n_y - 2}$$

 $\sim t_{n_x + n_y - 2}$

$$\rightarrow CI_{1-\alpha}(\mu_x - \mu_y) = \bar{x} - \bar{y} \pm t_{n_x + n_y - 2} \sqrt{\frac{s_{\text{pool}}^2}{n_x} + \frac{s_{\text{pool}}^2}{n_y}}$$

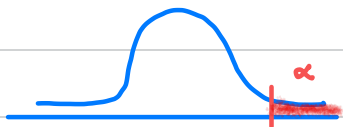
confidence interval
for $\mu_x - \mu_y$,
unknown equal
variances $\sigma_x^2 = \sigma_y^2$

Ⓟ LI.diff.mean (x, y, type = "independent", sigma.x, sigma.y, conf.level)

PERVS XXX : PERVS 3

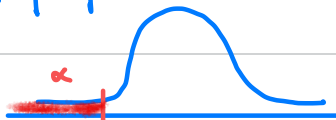
$$H_0: \mu_x - \mu_y \leq \delta_0 \quad (= \delta_0)$$

$$H_1: \mu_x - \mu_y > \delta_0 \quad (= \delta_1 > \delta_0)$$



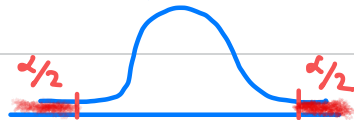
$$H_0: \mu_x - \mu_y \geq \delta_0 \quad (= \delta_0)$$

$$H_1: \mu_x - \mu_y < \delta_0 \quad (= \delta_1 < \delta_0)$$



$$H_0: \mu_x - \mu_y = \delta_0$$

$$H_1: \mu_x - \mu_y \neq \delta_0$$



KNOWN VARIANCES
 σ_x^2, σ_y^2

$$\frac{(\bar{x} - \bar{y}) - \delta_0}{\sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}} \stackrel{H_0}{\sim} N(0,1) \quad \text{PIVOTAL QUANTITY}$$

$$Z\text{-score} = \frac{(\bar{x} - \bar{y}) - \delta_0}{\sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}} \quad \text{REALISATION OF PIVOTAL QUANTITY}$$

$$R_\alpha = \{Z\text{-score} > \boxed{Z_\alpha}\}$$

$qnorm(1-\alpha)$

$$p\text{-value} = P(Z > Z\text{-score})$$

$\hookrightarrow 1 - pnorm(Z\text{-score})$

$$R_\alpha = \{Z\text{-score} < \boxed{-Z_\alpha}\}$$

$qnorm(\alpha)$

$$p\text{-value} = P(Z < Z\text{-score})$$

$\hookrightarrow pnorm(Z\text{-score})$

$$R_\alpha = \{Z\text{-score} < \boxed{-Z_{\alpha/2}} \text{ or } Z\text{-score} > \boxed{Z_{\alpha/2}}\}$$

$qnorm(1-\alpha/2)$

$$p\text{-value} = 2 \cdot P(Z > |Z\text{-score}|)$$

$\hookrightarrow [1 - pnorm(|Z\text{-score}|)] \cdot 2$

UNKNOWN VARIANCES
ASSUME EQUAL $\sigma_x^2 = \sigma_y^2$

$$\frac{(\bar{x} - \bar{y}) - \delta_0}{\sqrt{\frac{s_{pooled}^2}{n_x} + \frac{s_{pooled}^2}{n_y}}} \stackrel{H_0}{\sim} t_{n_x+n_y-2} \quad \text{PIVOTAL QUANTITY}$$

$$t\text{-score} = \frac{(\bar{x} - \bar{y}) - \delta_0}{\sqrt{\frac{s_{pooled}^2}{n_x} + \frac{s_{pooled}^2}{n_y}}} \quad \text{REALISATION OF PIVOTAL QUANTITY}$$

$$R_\alpha = \{t\text{-score} > \boxed{t_{n_x+n_y-2, \alpha}}\}$$

$qt(1-\alpha, n_x+n_y-2)$

$$p\text{-value} = P(t_{n_x+n_y-2} > t\text{-score})$$

$\hookrightarrow 1 - pt(t\text{-score}, n_x+n_y-2)$

$$R_\alpha = \{t\text{-score} < \boxed{-t_{n_x+n_y-2, \alpha}}\}$$

$qt(\alpha, n_x+n_y-2)$

$$p\text{-value} = P(t_{n_x+n_y-2} < t\text{-score})$$

$\hookrightarrow pt(t\text{-score}, n_x+n_y-2)$

$$R_\alpha = \{t\text{-score} < \boxed{-t_{n_x+n_y-2, \frac{\alpha}{2}}} \text{ or } t\text{-score} > \boxed{t_{n_x+n_y-2, \frac{\alpha}{2}}}\}$$

$qt(1-\frac{\alpha}{2}, n_x+n_y-2)$

$$p\text{-value} = 2 \cdot P(t_{n_x+n_y-2} > |t\text{-score}|)$$

$\hookrightarrow [1 - pt(|t\text{-score}|, n_x+n_y-2)] \cdot 2$

CASE	θ	$\hat{\theta}$	se: standard error estimate
$(x_1, y_1), \dots, (x_n, y_n)$ $D_1, D_2, \dots, D_n$ i.i.d. $E[D] = \mu_x - \mu_y$ $\text{Var}(D) = \sigma_x^2 + \sigma_y^2 - 2\sigma_{xy}$ $= \sigma_d^2$ X,Y PAIRED $D_i = x_i - y_i$	4 $\mu_x - \mu_y$ (μ_d)	$\bar{D} = \bar{X} - \bar{Y} \sim N\left(\mu_x - \mu_y, \frac{\sigma_d^2}{n}\right)$ $\bar{x} - \bar{y}$ $(\bar{d})$	$\frac{s_d}{\sqrt{n}}$ recall s_d is calculated: ① as standard deviation of the differences ② with $s_d^2 = s_x^2 + s_y^2 - 2s_{xy}$

$$\frac{\bar{D} - \mu_d}{\sigma_d / \sqrt{n}} \sim N(0,1) \longrightarrow \text{CI}_{1-\alpha}(\mu_d) = \bar{d} \pm Z_{\alpha/2} \frac{\sigma_d}{\sqrt{n}}$$

confidence interval for $\mu_d (= \mu_x - \mu_y)$, variance σ_d^2 known

$$\frac{\bar{D} - \mu_d}{s_d / \sqrt{n}} \sim t_{n-1} \longrightarrow \text{CI}_{1-\alpha}(\mu_d) = \bar{d} \pm t_{n-1, \alpha/2} \frac{s_d}{\sqrt{n}}$$

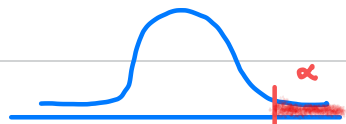
confidence interval for μ_d , variance unknown

② `ci.diffmean(x, y, type = "paired", sigma.d, conf.level)`

PEPVS XXX : PEPVS 4

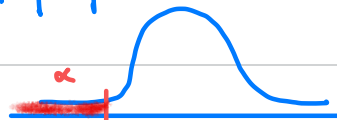
$$H_0: \mu_x - \mu_y \leq \delta_0 \quad (= \delta_0)$$

$$H_1: \mu_x - \mu_y > \delta_0 \quad (= \delta_1 > \delta_0)$$



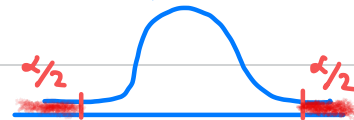
$$H_0: \mu_x - \mu_y \geq \delta_0 \quad (= \delta_0)$$

$$H_1: \mu_x - \mu_y < \delta_0 \quad (= \delta_1 < \delta_0)$$



$$H_0: \mu_x - \mu_y = \delta_0$$

$$H_1: \mu_x - \mu_y \neq \delta_0$$



KNOWN VARIANCE σ_D^2

$$\frac{(\bar{X} - \bar{Y}) - \delta_0}{\sigma_D / \sqrt{n}} \stackrel{H_0}{\sim} N(0,1) \quad \text{PIVOTAL QUANTITY}$$

$$Z\text{-score} = \frac{\bar{x} - \bar{y} - \delta_0}{\sigma_D / \sqrt{n}} \quad \text{REALISATION OF PIVOTAL QUANTITY}$$

$$R_\alpha = \{Z\text{-score} > \boxed{z_\alpha}\}$$

$q_{\text{norm}}(1-\alpha) \leftarrow$

$$\text{p-value} = P(Z > Z\text{-score})$$

$\leftarrow 1 - \text{pnorm}(Z\text{-score})$

$$R_\alpha = \{Z\text{-score} < \boxed{-z_\alpha}\}$$

$q_{\text{norm}}(\alpha) \leftarrow$

$$\text{p-value} = P(Z < Z\text{-score})$$

$\leftarrow \text{pnorm}(Z\text{-score})$

$$R_\alpha = \{Z\text{-score} < \boxed{-z_{\alpha/2}} \text{ or } Z\text{-score} > \boxed{z_{\alpha/2}}\}$$

$q_{\text{norm}}(1-\alpha/2) \leftarrow$

$$\text{p-value} = 2 \cdot P(Z > |Z\text{-score}|)$$

$\leftarrow [1 - \text{pnorm}(|Z\text{-score}|)] \cdot 2$

UNKNOWN VARIANCES

$$\frac{\bar{X} - \bar{Y} - \delta_0}{S_D / \sqrt{n}} \stackrel{H_0}{\sim} t_{n-1} \quad \text{PIVOTAL QUANTITY}$$

$$t\text{-score} = \frac{\bar{x} - \bar{y} - \delta_0}{s_D / \sqrt{n}} \quad \text{REALISATION OF PIVOTAL QUANTITY}$$

$$R_\alpha = \{t\text{-score} > \boxed{t_{n-1, \alpha}}\}$$

$q_{t(1-\alpha, n-1)} \leftarrow$

$$\text{p-value} = P(t_{n-1} > t\text{-score})$$

$\leftarrow 1 - \text{pt}(t\text{-score}, n-1)$

$$R_\alpha = \{t\text{-score} < \boxed{-t_{n-1, \alpha}}\}$$

$q_{t(\alpha, n-1)} \leftarrow$

$$\text{p-value} = P(t_{n-1} < t\text{-score})$$

$\leftarrow \text{pt}(t\text{-score}, n-1)$

$$R_\alpha = \{t\text{-score} < \boxed{-t_{n-1, \frac{\alpha}{2}}} \text{ or } t\text{-score} > \boxed{t_{n-1, \frac{\alpha}{2}}}\}$$

$q_{t(1-\frac{\alpha}{2}, n-1)} \leftarrow$

$$\text{p-value} = 2 \cdot P(t_{n-1} > |t\text{-score}|)$$

$\leftarrow [1 - \text{pt}(|t\text{-score}|, n-1)] \cdot 2$

PEPUS POINT ESTIMATION POWERFUL ULTIMATE SENTENCE (PEPUS)

CASES

$X_1, \dots, X_{n_x}$ i.i.d.
 $X_i \sim \text{Bernoulli}(p_x)$
 $Y_1, \dots, Y_{n_y}$ i.i.d.
 $Y_i \sim \text{Bernoulli}(p_y)$
 X, Y independent

θ

θ

$p_x - p_y$

$\hat{\theta}$

$\hat{p}_x - \hat{p}_y$

$$N(p_x - p_y, \frac{p_x(1-p_x)}{n_x} + \frac{p_y(1-p_y)}{n_y})$$

$\hat{\theta}$

$\hat{p}_x - \hat{p}_y$

se: standard error estimate

$$\sqrt{\frac{\hat{p}_x(1-\hat{p}_x)}{n_x} + \frac{\hat{p}_y(1-\hat{p}_y)}{n_y}}$$

$$\frac{\hat{p}_x - \hat{p}_y - (p_x - p_y)}{\sqrt{\frac{\hat{p}_x(1-\hat{p}_x)}{n_x} + \frac{\hat{p}_y(1-\hat{p}_y)}{n_y}}} \sim N(0,1)$$

$\sim N(0,1)$

$$ci_{1-\alpha}(p_x - p_y) = \hat{p}_x - \hat{p}_y \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_x(1-\hat{p}_x)}{n_x} + \frac{\hat{p}_y(1-\hat{p}_y)}{n_y}}$$

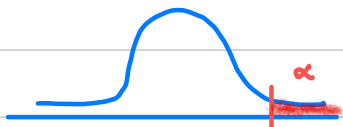
Confidence interval for $p_x - p_y$
 (variance always unknown)

Ⓡ `ci.diffprop(x, y, success.x, success.y, conf.level)`

PERVS XXX : PERVS J

$$H_0: p_x - p_y \leq 0 \quad (=0)$$

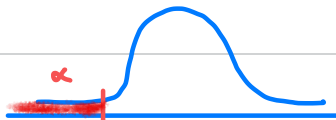
$$H_1: p_x - p_y > 0 \quad (= \delta_1 > 0)$$



$$\hat{p}_x = \frac{\# \text{ successes in } x}{n_x}$$

$$H_0: p_x - p_y \geq 0 \quad (=0)$$

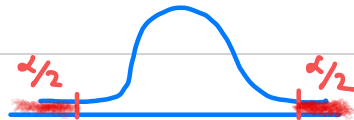
$$H_1: p_x - p_y < 0 \quad (= \delta_1 < 0)$$



$$\hat{p}_y = \frac{\# \text{ successes in } y}{n_y}$$

$$H_0: p_x = p_y$$

$$H_1: p_x \neq p_y$$



$$\hat{p}_0 = \frac{\# \text{ total successes}}{n_x + n_y} = \frac{n_x \hat{p}_x + n_y \hat{p}_y}{n_x + n_y}$$

$$\frac{\hat{p}_x - \hat{p}_y - 0}{\sqrt{\frac{\hat{p}_0(1-\hat{p}_0)}{n_x} + \frac{\hat{p}_0(1-\hat{p}_0)}{n_y}}} \stackrel{H_0}{\sim} N(0,1)$$

PIVOTAL
QUANTITY

$$\text{z-score} = \frac{\hat{p}_x - \hat{p}_y - 0}{\sqrt{\frac{\hat{p}_0(1-\hat{p}_0)}{n_x} + \frac{\hat{p}_0(1-\hat{p}_0)}{n_y}}}$$

REALISATION OF
PIVOTAL QUANTITY

$$R_\alpha = \{ \text{z-score} > \boxed{z_\alpha} \}$$

$q_{\text{norm}}(1-\alpha)$

$$\text{p-value} = P(Z > \text{z-score})$$

$\hookrightarrow 1 - \text{pnorm}(\text{z-score})$

$$R_\alpha = \{ \text{z-score} < \boxed{-z_\alpha} \}$$

$q_{\text{norm}}(\alpha)$

$$\text{p-value} = P(Z < \text{z-score})$$

$\hookrightarrow \text{pnorm}(\text{z-score})$

$$R_\alpha = \{ \text{z-score} < -z_{\alpha/2} \text{ or } \text{z-score} > \boxed{z_{\alpha/2}} \}$$

$q_{\text{norm}}(1-\alpha/2)$

$$\text{p-value} = 2 \cdot P(Z > |\text{z-score}|)$$

$\hookrightarrow [1 - \text{pnorm}(|\text{z-score}|)] \cdot 2$